

Machine Learning Prediction of Antimicrobial Susceptibility

(S vs non-S) from Quality-Controlled Hospital Microbiology Data

**Jittiphat Sanom¹, Chanchanok Aramrat¹, Kittikorn Hantrakul¹, Part
Pramokchon¹, Alongkot Gongmanee¹, and Somnuk Sinthupuan^{1*}**

**¹Department of Family Medicine, Faculty of Medicine, Chiang Mai University,
Chiang Mai 50200, Thailand**

***Corresponding author, Email address: somnuk@mju.ac.th**

Abstract

Hospital microbiology data are frequently affected by missing values, duplicate records, inconsistent naming, and heterogeneous coding across information systems, all of which limit their suitability for machine-learning analysis. This study developed a machine-learning framework to predict antimicrobial susceptibility of microorganisms, classified as sensitive (S) versus non-sensitive (non-S; combining intermediate and resistant), using patient-context features after systematic data quality assessment.

An end-to-end pipeline was designed to evaluate data quality in two dimensions, completeness and consistency, followed by preprocessing to standardize organism names, specimen types, antibiotic names, and susceptibility-result fields using dictionary mapping, regular expressions, and fuzzy matching. Intrinsic-resistance rules were incorporated, records that could not be interpreted reliably were flagged as REVIEW and excluded from supervised learning, and repeated isolates were handled using a study-defined 30-day first-isolate rule to reduce bias from duplicate results.

After all preprocessing and eligibility criteria, including the rolling 30-day first-isolate rule, were applied, 305,282 records remained for model development. Four tree-based machine-learning models—Decision Tree, Random Forest, XGBoost, and CatBoost—were trained to predict sensitive (S) versus non-sensitive (non-S) outcomes from microbiology and patient-context features. To mitigate class imbalance, CTGAN-based augmentation was applied exclusively to the training data after train-test splitting to balance the binary classes.

Model performance was evaluated using accuracy, precision, recall, and F1-score under three experimental settings with sequential feature removal. In the full-feature setting, which included organism, antibiotic, specimen, sex, age, and ward information, Random Forest achieved the highest accuracy of 77.23% with a macro F1-score of 0.7247. After organism information was removed, XGBoost achieved the best performance, with an accuracy of 67.67% and a macro F1-score of 0.6294. When both organism and ward information were removed, XGBoost again achieved the highest performance, with an accuracy of 67.21% and a macro F1-score of 0.6257.

These findings indicate that systematically quality-controlled hospital microbiology data can support machine-learning prediction of antimicrobial susceptibility and that organism information contributes substantially to predictive performance, while reduced feature sets still retained moderate predictive ability.

Keywords: data quality assessment; data preprocessing; machine learning; antimicrobial susceptibility testing; binary classification; antimicrobial susceptibility prediction; antimicrobial resistance; CTGAN; tree-based models.

1. Introduction

Culture and antimicrobial susceptibility testing (AST) data are fundamental for the treatment of infectious diseases, antimicrobial stewardship, and antimicrobial resistance (AMR) surveillance in hospitals. Cumulative antimicrobial susceptibility reports, or antibiograms, are widely used to summarize institutional susceptibility patterns and support empiric antimicrobial selection. The CLSI M39 guideline provides an important framework for the analysis and presentation of cumulative antimicrobial susceptibility test data so that such reports are reliable and clinically useful (Clinical and Laboratory Standards Institute [CLSI], 2022a).

In practice, however, hospital microbiology data are often affected by multiple data quality problems, including missing values, duplicate records, non-standard organism and antibiotic names, inconsistent specimen labels, and heterogeneous coding across information systems. These problems reduce not only the reliability of cumulative susceptibility reporting but also the readiness of the data for machine-learning analysis. Systems such as WHONET and BacLink were developed to facilitate microbiology data management and standardization, highlighting the importance of data quality assessment and preprocessing before advanced analysis (Stelling et al., 2007; WHONET, 2026).

In recent years, machine learning has increasingly been applied to antimicrobial resistance and susceptibility prediction. Reviews of the literature have shown that tree-based models, particularly Random Forest and gradient-boosted trees, often perform well with real-world healthcare data because they can capture nonlinear relationships and accommodate complex interactions among variables (Anahtar et al., 2021; Ardila et al., 2025; Sakagianni et al., 2023). Nevertheless, many AMR prediction studies still face

common limitations, including poor source-data quality, class imbalance, inconsistent preprocessing pipelines, and limited discussion of how categorical AST labels were derived and standardized (Ardila et al., 2025; Sakagianni et al., 2023).

Accordingly, this study aimed to develop a machine-learning framework for predicting antimicrobial susceptibility under a binary classification setting (S vs non-S) from hospital microbiology data after systematic data quality assessment and preprocessing. The specific objectives were: (1) to assess and improve the quality of hospital microbiology data in terms of completeness and consistency before model development; (2) to develop a preprocessing workflow that standardizes organism names, antibiotic names, specimen types, and susceptibility results into a machine-learning-ready format; (3) to develop and compare the predictive performance of multiple machine-learning models for binary antimicrobial susceptibility classification (S vs non-S); and (4) to examine the contribution of microbiology variables and patient-context variables to predictive performance.

2. Related Work

Relevant studies can be grouped into four major areas: (1) surveillance systems and cumulative antimicrobial susceptibility reporting, (2) standardization and integration of microbiology data, (3) machine-learning applications for antimicrobial susceptibility prediction, and (4) methodological gaps related to data quality, class imbalance, and real-world implementation.

In the area of surveillance and reporting, CLSI M39 provides an important framework for the analysis and presentation of cumulative antimicrobial susceptibility

test data to support clinical decision-making and antimicrobial stewardship (CLSI, 2022a). At the global level, the World Health Organization established the Global Antimicrobial Resistance and Use Surveillance System (GLASS) to standardize the collection, analysis, and sharing of AMR data (World Health Organization, 2022, 2023). In Thailand, the National Antimicrobial Resistance Surveillance Center of Thailand publishes annual antibiogram reports and surveillance summaries, underscoring the direct role of AST data in AMR monitoring and antibiotic stewardship (National Antimicrobial Resistance Surveillance Center of Thailand, 2024, 2025).

In the area of standardization and microbiology data integration, several studies have shown that the major challenge in hospital microbiology datasets is not simply the absence of data but also inconsistency of formats, nonuniform coding, and fragmentation across information systems. WHONET and BacLink were developed to support the capture, conversion, processing, and analysis of microbiology data from legacy systems, particularly for surveillance use (Stelling et al., 2007; WHONET, 2026). In the Thai context, previous work has demonstrated ETL-based processing and visualization pipelines for bacterial profiles and antibiotic susceptibility profiles from hospital data, emphasizing data cleaning, quality assessment, and bias reduction before further analysis (Aramrat & Boonma, 2023).

For machine-learning applications, previous studies have demonstrated that ML can support antimicrobial stewardship, resistance phenotype prediction, and clinical decision support. Personalized antibiogram approaches based on electronic health records have shown the ability to predict antibiotic susceptibility and support narrower-spectrum therapy while maintaining adequate coverage (Corbin et al., 2022). Multitask learning frameworks have also been proposed to predict individualized resistance probabilities

across multiple antimicrobial classes (Goto et al., 2026). Collectively, these studies suggest that real-world clinical and microbiology data can be used for clinically meaningful susceptibility prediction (Anahtar et al., 2021; Corbin et al., 2022; Goto et al., 2026).

Despite these advances, major challenges remain. Recent reviews have emphasized that model performance in AMR prediction depends not only on algorithm choice but also on the quality and consistency of the underlying data, the handling of class imbalance, and the rigor of evaluation procedures (Ardila et al., 2025; Sakagianni et al., 2023). These limitations motivate the present study, which directly focuses on the integration of data quality assessment, systematic preprocessing, and binary antimicrobial susceptibility prediction (S vs non-S) using real-world hospital microbiology data.

3. Materials and Methods

3.1. Study Design and Data Source

This retrospective study used microbiology and anonymized patient-context data from Chiang Mai University hospital, a tertiary hospital in Chiang Mai province, Thailand, collected from 8 September 2011 to 2 January 2020. A total of 1,097,459 inpatient culture records were initially obtained from the IPD Culture database. Data quality assessment was performed for key microbiology and patient-context variables, including organism name, antibiotic name, specimen type, susceptibility result, sex, age, and ward of admission.

The records then underwent sequential preprocessing to clean, standardize, and transform the data. To ensure data quality and clinical relevance, the **eligibility criteria**

for retaining records were defined as follows: (1) records must originate from inpatient (IPD) cultures and successfully merge with the patient-context database via admission numbers; (2) susceptibility outcomes must be available and subsequently transformed into a binary classification (Sensitive [S] vs non-sensitive [non-S], where non-S combines Intermediate [I] and Resistant [R]); (3) records must pass the data quality assessment, excluding any entries flagged as 'REVIEW' due to unresolvable non-standard entries or missing values; and (4) inclusion was restricted to the first isolate per patient, per organism, per antibiotic within a 30-day period (the 30-day first-isolate rule) to mitigate bias from repeated sampling. After all preprocessing steps, eligibility criteria, and the rolling 30-day first-isolate rule were applied, a final cohort of 305,282 records remained for model development.

3.2. Research Ethics

This study was reviewed and approved by the Human Research Ethics Committee, Faculty of Medicine, Chiang Mai University, Committee 5 (Research ID: 8889; Study Code: FAM-2565-08889). The present study is a secondary analysis of data derived from a previously approved research project. It involved retrospective hospital microbiology and clinical record data, with no direct patient contact or intervention. All data were de-identified prior to analysis. The requirement for individual informed consent was waived by the ethics committee because the study posed minimal risk and did not adversely affect the rights or welfare of the patients.

3.3. Data Quality Assessment

This study assessed data quality in two primary dimensions, namely completeness and consistency, because both dimensions directly affect data readiness for machine

learning. Let the full dataset be denoted by $D = \{x_1, x_2, \dots, x_N\}$, where N is the total number of records and x_{iv} is the value of variable v in record i .

The completeness of variable v was defined as:

$$\text{Completeness}(v) = \frac{1}{N} \sum_{i=1}^N I(x_{iv} \text{ is not missing}),$$

where $I(\cdot)$ is the indicator function.

The consistency of variable v relative to a predefined standard-value set S_v was defined as:

$$\text{Consistency}(v) = \frac{1}{N} \sum_{i=1}^N I(x_{iv} \in S_v).$$

These measures were used to identify fields with missing values, non-standard naming, or inconsistent data formats before the preprocessing stage.

3.4. Data Preprocessing

The preprocessing workflow was designed for microbiology data originating from multiple sources and containing inconsistent naming conventions. It consisted of five main steps.

3.4.1. Data Standardization

Let r denote a raw value, such as an organism name, antibiotic name, specimen type, or susceptibility-result string, and let $T = \{t_1, t_2, \dots, t_m\}$ denote the corresponding set of standardized terms. The standardization function $f(r)$ was defined as:

$$f(r) = \begin{cases} \text{dictionary match,} & \text{if } r \text{ matches a predefined mapping} \\ \text{regex-normalized match,} & \text{if } r \text{ matches a standard pattern} \\ \arg \max_{t \in T} \text{sim}(r, t), & \text{if } \max_{t \in T} \text{sim}(r, t) \geq \tau \\ \text{REVIEW,} & \text{otherwise,} \end{cases}$$

where $\text{sim}(r, t)$ is the similarity score between raw value r and standardized term t , and τ is the fuzzy-matching threshold. This procedure was applied to organism names, antibiotic names, specimen types, and susceptibility-result fields.

3.4.2. Intrinsic-Resistance Checking

Let o_i denote the standardized organism for record i , a_i the standardized antibiotic, y_i the recorded susceptibility category, and R_{int} the rule base of organism-antibiotic pairs with known intrinsic resistance. The intrinsic-resistance inconsistency flag was defined as:

$$\text{IRFlag}_i = \begin{cases} 1, & \text{if } (o_i, a_i) \in R_{\text{int}} \text{ and } y_i = S \\ 0, & \text{otherwise.} \end{cases}$$

If $\text{IRFlag}_i = 1$, the record was flagged for review.

3.4.3. REVIEW Flag Assignment

Let status_i denote the final preprocessing status of record i . A record was assigned REVIEW status if it met at least one of the following conditions: unresolved organism mapping, unresolved antibiotic mapping, unresolved specimen mapping, invalid or ambiguous susceptibility-result coding, or intrinsic-resistance inconsistency.

$$\text{status}_i = \begin{cases} \text{REVIEW,} & \text{if the record cannot be reliably interpreted} \\ \text{KEEP,} & \text{otherwise.} \end{cases}$$

The supervised-learning dataset before duplicate reduction was therefore defined as:

$$D_{\text{sup}} = \{i: \text{status}_i = \text{KEEP and } y_i \in \{S, I, R\}\}.$$

3.4.4. First-Isolate Rule for Duplicate Reduction

Let each record contain a patient identifier, standardized organism name, standardized antibiotic name, and test date. To reduce repeated antimicrobial susceptibility results from the same patient, the first-isolate rule was applied using a rolling 30-day window at the patient–organism–antibiotic level.

$$F_i = \begin{cases} 1, & \text{if there is no earlier retained record } j \text{ such that } p_j = p_i, o_j = o_i, s_j = s_i, \\ & \text{and } 0 < d_i - d_j \leq 30 \text{ days} \\ 0, & \text{otherwise.} \end{cases}$$

For each patient–organism–antibiotic combination, records were sorted chronologically by test date. The first record was retained, and subsequent records were retained only if they occurred more than 30 days after the most recently retained record. Records occurring within 30 days of the most recently retained record for the same patient, organism, and antibiotic were treated as repeated results and excluded from the modeling dataset.

$$D_{\text{dedup}} = \{i \in D_{\text{sup}} : F_i = 1\}.$$

Specimen type and susceptibility result were not used as deduplication keys. This was done to avoid retaining repeated susceptibility results solely because they came from different specimen types or had different outcome labels.

3.4.5. Final Dataset for Modeling

The final analysis dataset was defined as:

$$D_{\text{final}} = D_{\text{dedup}}.$$

The explanatory feature vector for each record was:

$$X = [\text{organism}, \text{antibiotic}, \text{specimen}, \text{sex}, \text{age}, \text{ward}],$$

and The target variable was defined as a binary outcome:

Sensitive (S) versus non-Sensitive (non-S), where non-S combines Intermediate (I) and Resistant (R).

This transformation was applied prior to model training and was consistently used throughout all experiments.

The study used the final categorical AST interpretations recorded by the hospital laboratory as the basis for the prediction target. These interpretations were transformed into a binary outcome: Sensitive (S) versus non-sensitive (non-S), where non-S combines Intermediate (I) and Resistant (R). Because the study period spanned 2011 to 2020, breakpoint criteria may have varied over time. The study did not retrospectively recalculate AST categories using a single breakpoint version across the full study period; instead, it used the categorical interpretations recorded in the hospital dataset at the time of testing. For contextual reference, the study period broadly spans the CLSI M100 editions from M100-S21 (2011) to the 30th edition (2020) (CLSI, 2011, 2020). In this study, category I denotes Intermediate.

3.5. Handling Class Imbalance

Because the dataset was imbalanced under the binary classification setting (Sensitive [S] vs non-sensitive [non-S], where non-S combines Intermediate [I] and Resistant [R]), CTGAN (Conditional Tabular Generative Adversarial Network) was applied to generate synthetic samples for the minority class.

243 CTGAN was trained exclusively using the training dataset after train-test
244 splitting and was not applied to the validation or test sets.

245 Synthetic samples were generated for the minority class (non-S) until both
246 classes (S and non-S) reached equal sample sizes, ensuring a fully balanced binary class
247 distribution for model training.

248 This approach ensured balanced class distribution and reduced bias toward the
249 majority class during model training.

250

3.6. Experimental Design

The dataset was split into training and test sets using an 80:20 ratio, with stratified sampling to preserve the class distribution of the binary outcome (S vs non-S).

No separate validation set was used; instead, internal model evaluation was conducted using cross-validation within the training set. Stratified sampling was applied to preserve the class distribution of the binary outcome (S vs non-S) in both sets.

CTGAN-based augmentation was applied only to the training set before model training and after data splitting to balance the classes. The test set was kept unchanged to reflect real-world data distribution.

To evaluate the contribution of feature groups, three experimental settings were defined:

- Setting 1: organism + antibiotic + specimen + sex + age + ward
- Setting 2: antibiotic + specimen + sex + age + ward
- Setting 3: antibiotic + specimen + sex + age

This design allows assessment of the impact of microbiological and patient-context features on predictive performance.

3.7. Machine-Learning Models

This study compared four tree-based machine-learning models: Decision Tree, Random Forest, XGBoost, and CatBoost. Let x_i denote the feature vector of sample i . The final class prediction of each model can be expressed in general form as:

$$\hat{y}_i = \arg \max_{c \in \{S, I, R\}} P(y_i = c | x_i).$$

3.7.1. Decision Tree

Decision Tree selects splits by maximizing impurity reduction. In this study, Gini impurity was used:

$$G(t) = 1 - \sum_{c \in \{S, I, R\}} p(c | t)^2,$$

where $p(c | t)$ is the proportion of class c at node t . The impurity reduction obtained by split sat node t was defined as:

$$\Delta G(s, t) = G(t) - \frac{N_L}{N_t} G(t_L) - \frac{N_R}{N_t} G(t_R),$$

where N_t is the number of samples at node t , and N_L and N_R are the sample counts in the left and right child nodes, respectively.

3.7.2. Random Forest

Random Forest is an ensemble of multiple decision trees. The final prediction is obtained by majority vote:

$$\hat{y}_i = \text{mode}\{h_1(x_i), h_2(x_i), \dots, h_B(x_i)\},$$

where $h_b(x_i)$ is the class predicted by tree b , and B is the number of trees. The class probability can be written as:

$$P(y_i = c | x_i) = \frac{1}{B} \sum_{b=1}^B I(h_b(x_i) = c).$$

3.7.3. XGBoost

XGBoost is based on gradient boosting, in which trees are added sequentially to reduce residual error:

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), f_k \in \mathcal{F},$$

where $\mathcal{F}$ is the space of regression trees. The objective function is:

$$\mathcal{L} = \sum_{i=1}^n l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k),$$

where $l(\cdot)$ is the training loss and $\Omega(\cdot)$ is the regularization term.

3.7.4. CatBoost

CatBoost is a gradient-boosting algorithm designed specifically for categorical features and based on ordered boosting:

$$\hat{y}_i^{(m)} = \hat{y}_i^{(m-1)} + \eta f_m(x_i),$$

where f_m is the base learner at iteration m and η is the learning rate.

3.7.5. Model Configuration and Experimental Environment

Categorical variables, including standardized organism name, standardized antibiotic name, standardized specimen type, sex, and ward, were encoded using Label Encoding before being used in the Decision Tree, Random Forest, and XGBoost models. CatBoost handled categorical features directly without explicit numeric encoding.

Hyperparameters were not optimized using Grid Search or Random Search. Instead, manual tuning was applied. The Decision Tree model used `max_depth = 10`. Random Forest used `n_estimators = 100`. XGBoost used `n_estimators = 100`, `learning_rate = 0.1`, and `max_depth = 6`. CatBoost used `iterations = 100`, `learning_rate = 0.1`, and `depth = 6`. Macro F1-score was used as the primary model-selection criterion under the binary classification setting (S vs non-S).

To ensure reproducibility, a random seed of 42 was used in all steps involving randomness, including data splitting, shuffling after CTGAN-based augmentation, and model training for Decision Tree, Random Forest, XGBoost, and CatBoost.

All analyses were conducted using Python 3.12 with the following libraries: pandas 3.0.2, scikit-learn 1.8.0, xgboost 3.2.0, catboost 1.2.10, and ctgan 0.12.1.

3.8. Evaluation Metrics

Model performance was evaluated using accuracy, Precision, Recall, and F1-score, and macro F1-score. All evaluation metrics were computed under the binary classification setting (S vs non-S). Particular emphasis was placed on macro F1-score because the class distribution was imbalanced.

Accuracy was defined as:

$$\text{Accuracy} = \frac{1}{n} \sum_{i=1}^n I(\hat{y}_i = y_i).$$

For each class c , precision and recall were defined as:

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c},$$

$$\text{Recall}_c = \frac{TP_c}{TP_c + FN_c}.$$

The F1-score for class c was defined as:

$$F1_c = \frac{2 \times \text{Precision}_c \times \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}.$$

The macro F1-score was computed as:

$$\text{Macro F1} = \frac{1}{3} \sum_{c \in \{S, I, R\}} F1_c.$$

4. Results

4.1. Data Quality Assessment and Preprocessing Results

Before model development, an initial set of 1,097,459 inpatient culture records was retrieved from the IPD Culture database. Data quality was assessed in terms of completeness and consistency for key microbiological and patient-related variables. The overall completeness of the dataset was high; antibiotic names, specimen types, and susceptibility outcomes demonstrated 100.00% completeness, while organism names showed 96.22% completeness.

Following the quality assessment, the data underwent systematic preprocessing, including dictionary mapping, regular expression-based cleaning, and fuzzy matching techniques to standardize terminology and resolve inconsistencies. During this process, records with ambiguous or unresolvable entries were flagged as 'REVIEW'. A total of 30,562 records (approximately 4.75% of the filtered dataset) were excluded, primarily due to unresolved organism nomenclature or conflicts related to intrinsic resistance rules. This step ensured high label reliability for subsequent supervised learning.

The eligibility criteria for inclusion in this study required that the data: (1) originated from inpatient (IPD) cultures and were successfully merged with the clinical IPD dataset using admission numbers; (2) contained valid antimicrobial susceptibility results; (3) passed quality control by excluding entries flagged as 'REVIEW' or containing missing standardized nomenclature; and (4) followed a 30-day first-isolate rule per patient, organism, and antibiotic to reduce bias from repeated sampling. After applying all preprocessing workflows, eligibility criteria, and the rolling 30-day first-isolate rule, a total of 305,282 records remained for model development.

To align with clinical decision-making and model stability, the antimicrobial susceptibility outcome was subsequently reformulated into a binary classification problem. Specifically, “Sensitive (S)” was defined as the positive class, while “Intermediate (I)” and “Resistant (R)” were combined into a single “non-Sensitive (non-S)” class. This transformation facilitated clearer interpretation and improved model robustness in downstream analysis.

To further address class imbalance, CTGAN was applied only to the training dataset after train-test splitting to generate synthetic samples for the minority class. The augmentation process ensured that both classes (S and non-S) were equally represented within the training dataset used for machine-learning experiments.

4.2. Model Comparison Results

The comparison results of the four machine-learning models across the three experimental settings are presented in Table 1. Model performance was evaluated using accuracy, macro precision, macro recall, and macro F1-score under the binary classification setting (Sensitive [S] vs non-sensitive [non-S]).

As shown in Table 1, Random Forest achieved the best performance in Setting 1, which used the full feature set consisting of organism, antibiotic, specimen type, sex, age, and ward information. In this setting, Random Forest achieved the highest accuracy of 77.23%, with a macro precision of 0.7142, macro recall of 0.7449, and macro F1-score of 0.7247.

After organism information was removed in Setting 2, model performance decreased across all models. In this reduced-feature setting, XGBoost achieved the best performance, with an accuracy of 67.67%, macro precision of 0.6276, macro recall of 0.6572, and macro F1-score of 0.6294.

In Setting 3, where both organism and ward information were excluded and only antibiotic, specimen type, sex, and age were retained, XGBoost again achieved the highest performance, with an accuracy of 67.21%, macro precision of 0.6247, macro recall of 0.6543, and macro F1-score of 0.6257.

Overall, model performance declined after removal of organism information, indicating that organism identity contributed substantially to antimicrobial susceptibility prediction. However, the relatively similar performance between Settings 2 and 3

suggests that ward information provided only limited additional predictive value after organism information had been excluded.

Setting	Features	Model	Accuracy	Macro		
				Precision	Recall	F1
Setting 1	Org, Anti, Specimen, Sex, Age, Ward	Decision Tree	76.01%	0.70	0.74	0.71
		Random Forest	77.23%	0.71	0.74	0.72
		XGBoost	76.42%	0.71	0.74	0.72
		CatBoost	74.58%	0.69	0.73	0.70
Setting 2	Anti, Specimen, Sex, Age, Ward	Decision Tree	67.33%	0.62	0.65	0.62
		Random Forest	66.85%	0.61	0.63	0.61
		XGBoost	67.67%	0.62	0.65	0.62
		CatBoost	65.26%	0.62	0.65	0.61
Setting 3	Anti, Specimen, Sex, Age	Decision Tree	66.70%	0.62	0.64	0.62
		Random Forest	65.47%	0.61	0.64	0.61
		XGBoost	67.21%	0.62	0.65	0.62
		CatBoost	66.19%	0.62	0.65	0.61

Table 1. Prediction results for antimicrobial susceptibility under binary classification (S vs non-S) across the three experimental settings.

4.3. Effect of Feature Reduction

To evaluate the contribution of feature groups in predicting antimicrobial susceptibility under the binary classification setting (Sensitive [S] vs non-sensitive [non-S]), the best-performing model in each experimental setting was compared under progressively reduced feature sets, as shown in Table 2.

Setting	Features	Best Model	Accuracy	Macro			Accuracy
				Precision	Recall	F1	Difference
Setting 1	Org, Anti, Specimen, Sex, Age, Ward	Random Forest	77.23%	0.71	0.74	0.72	—
Setting 2	Anti, Specimen, Sex, Age, Ward	XGBoost	67.67%	0.62	0.65	0.62	-9.56%
Setting 3	Anti, Specimen, Sex, Age	XGBoost	67.21%	0.62	0.65	0.62	-10.02%

Table 2. Comparison of the best-performing model under sequential feature reduction in binary classification (S vs non-S).

Model performance declined substantially after organism information was removed. Accuracy decreased from 77.23% in Setting 1 to 67.67% in Setting 2, corresponding to a reduction of 9.56 percentage points. This finding indicates that organism identity contributed strongly to antimicrobial susceptibility prediction.

When ward information was further removed in Setting 3, accuracy decreased slightly from 67.67% to 67.21%. Compared with the full-feature setting, Setting 3 showed an overall accuracy reduction of 10.02 percentage points. The relatively small difference

between Settings 2 and 3 suggests that ward information provided limited additional predictive value after organism information had already been excluded.

Overall, the feature-reduction analysis demonstrates that organism information was the most influential removed feature group. Antibiotic and specimen information still retained moderate predictive ability, but simplified feature sets were insufficient to fully capture the complexity of antimicrobial susceptibility patterns.

4.4. Class-Level Performance of the Best Model

To further evaluate model behavior at the class level, the class-wise performance of the best-performing model in each experimental setting was analyzed under the binary classification setting (Sensitive [S] vs non-sensitive [non-S]), as shown in Table 3.

Setting	Best Model	Class	Precision	Recall	F1-score
Setting 1	Random Forest	S	0.88	0.80	0.84
		non-S	0.55	0.69	0.61
Setting 2	XGBoost	S	0.84	0.70	0.76
		non-S	0.42	0.62	0.50
Setting 3	XGBoost	S	0.84	0.69	0.76
		non-S	0.41	0.62	0.49

Table 3. Class-wise classification performance of the best-performing model in each setting under binary classification (S vs non-S).

The best-performing models showed stronger performance for the Sensitive (S) class than for the non-sensitive (non-S) class across all experimental settings. In Setting 1, Random Forest achieved an F1-score of 0.84 for the S class and 0.61 for the non-S class. After organism information was removed, XGBoost achieved non-S F1-scores of 0.50 in Setting 2 and 0.49 in Setting 3.

Although non-S precision was lower than S precision, non-S recall remained moderate across the three settings. This suggests that the models retained some ability to identify potentially intermediate or resistant isolates, although distinguishing non-S outcomes remained more challenging than identifying susceptible outcomes.

These findings indicate that organism information contributed substantially to class-level discrimination. After organism removal, model performance for the non-S class decreased, emphasizing the importance of pathogen identity in antimicrobial susceptibility prediction.

4.5. Effect of CTGAN-Based Class Balancing

To address class imbalance, CTGAN was applied exclusively to the training dataset under the binary classification setting (Sensitive [S] vs non-sensitive [non-S]). Before augmentation, the training dataset contained 244,225 records and showed class imbalance, with 180,962 records (74.10%) classified as S and 63,263 records (25.90%) classified as non-S.

CTGAN was trained only on the training subset after train-test splitting, and no synthetic samples were introduced into the test dataset. Synthetic samples were generated for the minority class (non-S) until both classes achieved equal representation within the training data.

After CTGAN-based augmentation, the training dataset increased to 361,924 records. Both S and non-S classes contained 180,962 records, corresponding to a balanced class distribution of 50.00% per class before model training.

Class	Before CTGAN in Training Set (n)	Before CTGAN (%)	After CTGAN in Training Set (n)	After CTGAN (%)
S	180,962	74.10	180,962	50.00
non-S	63,263	25.90	180,962	50.00
Total	244,225	100.00	361,924	100.00

Table 4. Distribution of classes in the training set before and after class balancing using CTGAN under binary classification (S vs non-S).

This training-only augmentation strategy reduced class imbalance while preserving an untouched real-world test set for model evaluation. As a result, model performance was assessed on real data that retained the original class distribution, improving the reliability of generalization assessment and reducing the risk of information leakage.

5. Discussion

This study developed a machine-learning framework for predicting antimicrobial susceptibility under a binary classification setting (Sensitive [S] versus non-sensitive [non-S]) using quality-controlled hospital microbiology data. The proposed framework incorporated systematic data quality assessment, preprocessing, a rolling 30-day first-isolate rule at the patient–organism–antibiotic level, CTGAN-based class balancing, and comparative machine-learning evaluation across progressively reduced feature settings.

After applying the rolling 30-day first-isolate rule, 305,282 records remained for model development. This rule reduced repeated antimicrobial susceptibility results from

the same patient, organism, and antibiotic within a 30-day window, thereby improving the clinical relevance of the modeling dataset and reducing bias from repeated observations.

Among the evaluated models, Random Forest achieved the best performance in the full-feature setting, which included organism, antibiotic, specimen type, sex, age, and ward information. In this setting, Random Forest achieved an accuracy of 77.23% and a macro F1-score of 0.7247. This result indicates that ensemble tree-based models can effectively capture complex relationships among microbiological features and patient-context variables when complete feature information is available.

After organism information was removed, XGBoost achieved the best performance in both reduced-feature settings. In Setting 2, which included antibiotic, specimen type, sex, age, and ward information, XGBoost achieved an accuracy of 67.67% and a macro F1-score of 0.6294. In Setting 3, which further excluded ward information and retained only antibiotic, specimen type, sex, and age, XGBoost achieved an accuracy of 67.21% and a macro F1-score of 0.6257. These findings suggest that XGBoost may be more robust than the other models when organism information is unavailable.

The feature-reduction analysis demonstrated that organism identity was an important contributor to antimicrobial susceptibility prediction. Model accuracy decreased from 77.23% in the full-feature setting to 67.67% after organism information was removed. In contrast, further removal of ward information resulted in only a small additional decrease in accuracy from 67.67% to 67.21%. This pattern suggests that organism information contributed substantially to predictive performance, whereas ward

information provided more limited additional value after organism information had already been excluded.

Class-level analysis showed that the best-performing models consistently achieved stronger performance for the Sensitive (S) class than for the non-sensitive (non-S) class. In the full-feature setting, Random Forest achieved an F1-score of 0.84 for the S class and 0.61 for the non-S class. In the reduced-feature settings, XGBoost achieved non-S F1-scores of 0.50 and 0.49 in Settings 2 and 3, respectively. These findings indicate that predicting non-S outcomes remained more challenging than predicting S outcomes, despite CTGAN-based class balancing.

The lower precision and F1-score for the non-S class may reflect the clinical and microbiological complexity of intermediate and resistant susceptibility patterns. Because the non-S class combined both Intermediate (I) and Resistant (R) categories, it likely represented a more heterogeneous outcome group than the S class. Nevertheless, the models maintained moderate recall for non-S outcomes, suggesting that the framework retained some ability to identify potentially intermediate or resistant isolates.

A key methodological strength of this study was the use of a leakage-safe CTGAN workflow. CTGAN-based augmentation was applied exclusively to the training dataset after train-test splitting, while the test set remained untouched and consisted only of real-world observations. This design reduced the risk of information leakage and allowed model performance to be evaluated under a more realistic class distribution.

Another important strength was the alignment between preprocessing logic and the study objective. The first-isolate rule was implemented using a rolling 30-day window

at the patient–organism–antibiotic level, rather than using technical duplicate identifiers such as barcode, specimen type, susceptibility result, or fixed date buckets. This approach better reflects the intended clinical purpose of reducing repeated susceptibility results from the same patient rather than merely removing technical duplicate records.

The study also highlights the importance of rigorous preprocessing and data quality control in hospital microbiology datasets. Dictionary mapping, regular expressions, fuzzy matching, intrinsic-resistance validation, REVIEW-based exclusion, and first-isolate filtering contributed to improving data consistency and label reliability before machine-learning analysis. These preprocessing steps are particularly important in real-world hospital datasets, where heterogeneous naming conventions, inconsistent coding, and repeated records can substantially affect downstream modeling results. From a clinical perspective, the findings suggest that quality-controlled microbiology data can support machine-learning prediction of antimicrobial susceptibility under a binary S vs non-S framework. However, the moderate performance in reduced-feature settings and the lower performance for non-S outcomes indicate that additional clinical variables, such as prior antibiotic exposure, comorbidities, previous culture history, and temporal resistance trends, may be needed to improve predictive performance for clinical deployment.

Overall, this study demonstrates that a carefully designed machine-learning pipeline using quality-controlled hospital microbiology data can provide clinically meaningful antimicrobial susceptibility prediction. The results emphasize the importance of organism information, the need for leakage-safe class-balancing procedures, and the

value of applying a clinically appropriate rolling 30-day first-isolate rule before model development.

5.1. Limitations

This study has several limitations. First, the dataset originated from a single tertiary-care hospital, which may limit generalizability to other healthcare settings with different pathogen distributions and antimicrobial usage patterns. Second, the retrospective design relied on routinely collected hospital microbiology data accumulated over multiple years, during which laboratory practices and susceptibility breakpoint criteria may have changed.

Third, although CTGAN-based augmentation improved class balance, synthetic data generation may not fully capture all real-world microbiological variability. In this study, CTGAN was applied exclusively to the training dataset after train-test splitting to minimize information leakage and improve reliability of model evaluation.

Fourth, external validation was not performed, and all models were evaluated using data from the same institution. Fifth, categorical variables were label-encoded for Decision Tree, Random Forest, and XGBoost models for computational convenience. Although tree-based models are generally less sensitive to ordinal encoding effects than linear models, some encoding-related bias may still remain.

Finally, the evaluation design used stratified train-test splitting to preserve class distribution and reduce imbalance bias; however, a fully temporal validation strategy for prospective clinical deployment was not performed.

Future studies should include multicenter external validation, temporal validation using future-year cohorts, and additional investigation of alternative encoding strategies,

advanced imbalance-handling techniques, explainable artificial intelligence approaches,
and clinically interpretable deployment frameworks.

6. Conclusion

This study developed a machine-learning framework for predicting antimicrobial susceptibility under a binary classification setting (Sensitive [S] vs non-sensitive [non-S]) using quality-controlled hospital microbiology data. The proposed framework integrated systematic data quality assessment, preprocessing, intrinsic-resistance checking, REVIEW-based exclusion, a rolling 30-day first-isolate rule, CTGAN-based class balancing, and comparative machine-learning evaluation across progressively reduced feature settings.

After applying the preprocessing workflow and the rolling 30-day first-isolate rule at the patient–organism–antibiotic level, 305,282 records remained for model development. CTGAN-based augmentation was applied exclusively to the training dataset after train-test splitting, while the test set was kept unchanged to provide a leakage-safe and realistic evaluation.

In the full-feature setting, which included organism, antibiotic, specimen type, sex, age, and ward information, Random Forest achieved the best performance, with an accuracy of 77.23% and a macro F1-score of 0.7247. After organism information was removed, XGBoost achieved the highest performance in the reduced-feature settings, with accuracies of 67.67% and 67.21% in Settings 2 and 3, respectively.

The findings demonstrate that organism information is an important contributor to antimicrobial susceptibility prediction. Removing organism information caused a

substantial reduction in predictive performance, whereas further removal of ward information resulted in only a small additional decrease. These results suggest that microbiological variables, particularly organism and antibiotic information, are central to susceptibility prediction, while reduced feature sets can still retain moderate predictive ability.

Overall, this study shows that quality-controlled hospital microbiology data can support machine-learning prediction of antimicrobial susceptibility under a binary S vs non-S framework. The proposed pipeline provides a foundation for future clinical decision-support systems and antimicrobial stewardship applications, although further external and temporal validation will be required before prospective clinical deployment.

References

- Anahtar, M. N., Yang, J. H., & Kanjilal, S. (2021). Applications of machine learning to the problem of antimicrobial resistance: An emerging model for translational research. *Journal of Clinical Microbiology*, 59(7), e01260-20.
- Aramrat, C., & Boonma, P. (2023). Development of data processing and visualization for bacterial and antibiotic susceptibility profile. *Data Science and Engineering Record*, 4(1).
- Ardila, C. M., González-Arroyave, D., & Tobón, S. (2025). Machine learning for predicting antimicrobial resistance in critical and high-priority pathogens: A systematic review considering antimicrobial susceptibility tests in real-world healthcare settings. *PLOS ONE*, 20(2), e0319460.
- Clinical and Laboratory Standards Institute. (2011). *Performance standards for*

598 *antimicrobial susceptibility testing* (21st informational supplement, CLSI
599 document M100-S21).

600 Clinical and Laboratory Standards Institute. (2020). *Performance standards for*
601 *antimicrobial susceptibility testing* (30th ed.; CLSI supplement M100).

602 Clinical and Laboratory Standards Institute. (2022a). *M39: Analysis and presentation of*
603 *cumulative antimicrobial susceptibility test data* (5th ed.).

604 Corbin, C. K., Sung, L., Chattopadhyay, A., Noshad, M., Chang, A., Deresinski, S.,
605 Baiocchi, M., & Chen, J. H. (2022). Personalized antibiograms for machine
606 learning-driven antibiotic selection. *Communications Medicine*, 2, Article 38.

607 Goto, M., Bandyopadhyay, A., Shi, Q., Wang, Y., Perencevich, E. N., Hernandez, D., &
608 Street, W. N. (2026). Personalized antibiogram: A novel multitask machine
609 learning framework for simultaneous prediction of antimicrobial resistance profile
610 with enhanced detection of carbapenem resistance in *Enterobacteriaceae*.
611 *Clinical Infectious Diseases*, ciag027. <https://doi.org/10.1093/cid/ciag027>

612 National Antimicrobial Resistance Surveillance Center of Thailand. (2024). *Annual*
613 *report of the National Antimicrobial Resistance Surveillance Center of Thailand*.

614 National Antimicrobial Resistance Surveillance Center of Thailand. (2025).
615 *Antibiograms by health service region*.

616 Sakagianni, A., Koufopoulou, C., Feretzakis, G., Kalles, D., Verykios, V. S., Myrianthefs,
617 P., & Fildisis, G. (2023). Using machine learning to predict antimicrobial
618 resistance: A literature review. *Antibiotics*, 12(3), 452.

619 Stelling, J. M., Kulldorff, M., & O'Brien, T. F. (2007). WHONET and BacLink: Software
620 tools for laboratory-based surveillance of infectious diseases and antimicrobial
621 resistance. *Advances in Disease Surveillance*, 2, 121.

622 WHONET. (2026). *WHONET microbiology laboratory database software* [Computer
623 software].

624 World Health Organization. (2022). *Global antimicrobial resistance and use surveillance*
625 *system (GLASS) report: 2022*.

626 World Health Organization. (2023). *Global antimicrobial resistance and use surveillance*
627 *system (GLASS)*.

628